QVI Chip Purchasing Behaviour Analysis
R Script

```
> setwd(C:/Users/Amooo/Videos/Quantium_Forage)
Error: unexpected '/' in "setwd(C:/"
> setwd("C:/Users/Amooo/Videos/Quantium_Forage")
> getwd()
[1] "C:/Users/Amooo/Videos/Quantium_Forage"
> install.packages("readxl")
Installing package into
‘C:/Users/Amooo/AppData/Local/R/win-library/4.5’
(as ‘lib’ is unspecified)
--- Please select a CRAN mirror for use in this session ---
also installing the dependencies ‘glue’, ‘utf8’, ‘rematch’, ‘cli’,
‘lifecycle’, ‘magrittr’, ‘pillar’, ‘pkgconfig’, ‘rlang’, ‘vctrs’,
‘crayon’, ‘hms’, ‘prettyunits’, ‘R6’, ‘cellranger’, ‘tibble’,
‘cpp11’, ‘progress’


  There is a binary version available but the source version is
later:
      binary source needs_compilation
vctrs  0.7.1  0.7.2              TRUE

  Binaries will be installed
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/glue_1.8.0.zi
p'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/utf8_1.2.6.zi
p'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/rematch_2.0.0
.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/cli_3.6.5.zip
'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/lifecycle_1.0
.5.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/magrittr_2.0.
4.zip'
```

```
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/pillar_1.11.1
.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/pkgconfig_2.0
.3.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/rlang_1.1.7.z
ip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/vctrs_0.7.1.z
ip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/crayon_1.5.3.
zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/hms_1.1.4.zip
'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/prettyunits_1
.2.0.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/R6_2.6.1.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/cellranger_1.
1.0.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/tibble_3.3.1.
zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/cpp11_0.5.3.z
ip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/progress_1.2.
3.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/readxl_1.4.5.
zip'
package ‘glue’ successfully unpacked and MD5 sums checked
package ‘utf8’ successfully unpacked and MD5 sums checked
package ‘rematch’ successfully unpacked and MD5 sums checked
package ‘cli’ successfully unpacked and MD5 sums checked
package ‘lifecycle’ successfully unpacked and MD5 sums checked
package ‘magrittr’ successfully unpacked and MD5 sums checked
package ‘pillar’ successfully unpacked and MD5 sums checked
package ‘pkgconfig’ successfully unpacked and MD5 sums checked
package ‘rlang’ successfully unpacked and MD5 sums checked
package ‘vctrs’ successfully unpacked and MD5 sums checked
```

```
package ‘crayon’ successfully unpacked and MD5 sums checked
package ‘hms’ successfully unpacked and MD5 sums checked
package ‘prettyunits’ successfully unpacked and MD5 sums checked
package ‘R6’ successfully unpacked and MD5 sums checked
package ‘cellranger’ successfully unpacked and MD5 sums checked
package ‘tibble’ successfully unpacked and MD5 sums checked
package ‘cpp11’ successfully unpacked and MD5 sums checked
package ‘progress’ successfully unpacked and MD5 sums checked
package ‘readxl’ successfully unpacked and MD5 sums checked

The downloaded binary packages are in

C:\Users\Amooo\AppData\Local\Temp\RtmpSGmHag\downloaded_packages
> install.packages("dplyr")
Installing package into
‘C:/Users/Amooo/AppData/Local/R/win-library/4.5’
(as ‘lib’ is unspecified)
also installing the dependencies ‘withr’, ‘generics’, ‘tidyselect’

trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/withr_3.0.2.z
ip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/generics_0.1.
4.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/tidyselect_1.
2.1.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/dplyr_1.2.0.z
ip'
package ‘withr’ successfully unpacked and MD5 sums checked
package ‘generics’ successfully unpacked and MD5 sums checked
package ‘tidyselect’ successfully unpacked and MD5 sums checked
package ‘dplyr’ successfully unpacked and MD5 sums checked

The downloaded binary packages are in

C:\Users\Amooo\AppData\Local\Temp\RtmpSGmHag\downloaded_packages
> install.packages("ggplot2")
Installing package into
‘C:/Users/Amooo/AppData/Local/R/win-library/4.5’
(as ‘lib’ is unspecified)
also installing the dependencies ‘farver’, ‘labeling’,
‘RColorBrewer’, ‘viridisLite’, ‘gtable’, ‘isoband’, ‘S7’, ‘scales’
```

```
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/farver_2.1.2.
zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/labeling_0.4.
3.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/RColorBrewer_
1.1-3.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/viridisLite_0
.4.3.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/gtable_0.3.6.
zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/isoband_0.3.0
.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/S7_0.2.1.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/scales_1.4.0.
zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/ggplot2_4.0.2
.zip'
package ‘farver’ successfully unpacked and MD5 sums checked
package ‘labeling’ successfully unpacked and MD5 sums checked
package ‘RColorBrewer’ successfully unpacked and MD5 sums checked
package ‘viridisLite’ successfully unpacked and MD5 sums checked
package ‘gtable’ successfully unpacked and MD5 sums checked
package ‘isoband’ successfully unpacked and MD5 sums checked
package ‘S7’ successfully unpacked and MD5 sums checked
package ‘scales’ successfully unpacked and MD5 sums checked
package ‘ggplot2’ successfully unpacked and MD5 sums checked

The downloaded binary packages are in

C:\Users\Amooo\AppData\Local\Temp\RtmpSGmHag\downloaded_packages
> install.packages("stringr")
Installing package into
‘C:/Users/Amooo/AppData/Local/R/win-library/4.5’
(as ‘lib’ is unspecified)
also installing the dependency ‘stringi’

trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/stringi_1.8.7
.zip'
```

```
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/stringr_1.6.0
.zip'
package ‘stringi’ successfully unpacked and MD5 sums checked
package ‘stringr’ successfully unpacked and MD5 sums checked

The downloaded binary packages are in

C:\Users\Amooo\AppData\Local\Temp\RtmpSGmHag\downloaded_packages
> install.packages("lubridate")
Installing package into
‘C:/Users/Amooo/AppData/Local/R/win-library/4.5’
(as ‘lib’ is unspecified)
also installing the dependency ‘timechange’

trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/timechange_0.
4.0.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/lubridate_1.9
.5.zip'
package ‘timechange’ successfully unpacked and MD5 sums checked
package ‘lubridate’ successfully unpacked and MD5 sums checked

The downloaded binary packages are in

C:\Users\Amooo\AppData\Local\Temp\RtmpSGmHag\downloaded_packages
> library(readxl)
Warning message:
package ‘readxl’ was built under R version 4.5.3
> library(dplyr)

Attaching package: ‘dplyr’

The following objects are masked from ‘package:stats’:

    filter, lag

The following objects are masked from ‘package:base’:

    intersect, setdiff, setequal, union

Warning message:
package ‘dplyr’ was built under R version 4.5.2
> library(ggplot2)
Warning message:
package ‘ggplot2’ was built under R version 4.5.3
> library(stringr)
```

```
Warning message:
package ‘stringr’ was built under R version 4.5.3
> library(lubridate)

Attaching package: ‘lubridate’

The following objects are masked from ‘package:base’:

    date, intersect, setdiff, union

Warning message:
package ‘lubridate’ was built under R version 4.5.3
> # ---------------------------------------------------
> # 1. LOAD THE DATA
> # ---------------------------------------------------
> # Load the transaction and purchase behaviour data
> # Make sure the file names are EXACTLY correct
> transaction_data <- read_excel("QVI Transaction Data.xlsx")
Error: `path` does not exist: ‘QVI Transaction Data.xlsx’
> purchase_behaviour <- read_excel("QVI Purchase Behaviour.xlsx")
Error: `path` does not exist: ‘QVI Purchase Behaviour.xlsx’
>
> # ---------------------------------------------------
> # 2. INITIAL DATA EXPLORATION & CLEANING
> # ---------------------------------------------------
> # Look at the first few rows of each dataset
> head(transaction_data)
Error: object 'transaction_data' not found
> head(purchase_behaviour)
Error: object 'purchase_behaviour' not found
>
> # Check the structure (column names and types)
> str(transaction_data)
Error: object 'transaction_data' not found
> str(purchase_behaviour)
Error: object 'purchase_behaviour' not found
>
> # Check for missing values
> colSums(is.na(transaction_data))
Error: object 'transaction_data' not found
> colSums(is.na(purchase_behaviour))
Error: object 'purchase_behaviour' not found
>
> # Check for and remove any duplicate transactions
> cat("Number of duplicate rows:",
sum(duplicated(transaction_data)), "\n")
Error: object 'transaction_data' not found
```

```
> transaction_data <- distinct(transaction_data) # Removes
duplicates
Error: object 'transaction_data' not found
>
> # Convert the DATE column from Excel's number format to a proper
date
> transaction_data$DATE <- as.Date(transaction_data$DATE, origin =
"1899-12-30")
Error: object 'transaction_data' not found
>
> # ---------------------------------------------------
> # 3. FEATURE ENGINEERING
> # ---------------------------------------------------
> # Extract pack size (the numbers) from the product name
> transaction_data$PACK_SIZE <-
as.numeric(str_extract(transaction_data$PROD_NAME, "\\d+"))
Error: object 'transaction_data' not found
>
> # Extract the first word of the product name as the brand
> transaction_data$BRAND <- word(transaction_data$PROD_NAME, 1)
Error: object 'transaction_data' not found
>
> # Clean up and standardize some brand names
> transaction_data <- transaction_data %>%
+   mutate(BRAND = case_when(
+     BRAND == "Red" ~ "RRD",
+     BRAND == "Snbts" ~ "Sunbites",
+     BRAND == "Infzns" ~ "Infuzions",
+     BRAND == "WW" ~ "Woolworths",
+     BRAND == "Smith" ~ "Smiths",
+     BRAND == "Dorito" ~ "Doritos",
+     BRAND == "GrnWves" ~ "Grain Waves",
+     BRAND == "Natural" | BRAND == "NCC" ~ "Natural Chip Co",
+     TRUE ~ BRAND
+   ))
Error: object 'transaction_data' not found
>
> # ---------------------------------------------------
> # 4. MERGE THE DATASETS
> # ---------------------------------------------------
> # Combine transaction data with customer data
> merged_data <- transaction_data %>%
+   left_join(purchase_behaviour, by = "LYLTY_CARD_NBR")
Error: object 'transaction_data' not found
>
> # ---------------------------------------------------
> # 5. ANALYZE CUSTOMER SEGMENTS
> # ---------------------------------------------------
```

```
> # Calculate key metrics for each customer segment
> segment_analysis <- merged_data %>%
+   group_by(LIFESTAGE, PREMIUM_CUSTOMER) %>%
+   summarise(
+     total_customers = n_distinct(LYLTY_CARD_NBR),
+     total_sales = sum(TOT_SALES),
+     total_transactions = n(),
+     avg_transactions_per_customer = total_transactions /
total_customers,
+     avg_sales_per_customer = total_sales / total_customers,
+     .groups = "drop"
+   ) %>%
+   arrange(desc(total_sales))
Error: object 'merged_data' not found
>
> # Display the top segments by total sales
> print("Top Customer Segments by Total Sales:")
[1] "Top Customer Segments by Total Sales:"
> print(head(segment_analysis, 10))
Error: object 'segment_analysis' not found
>
> # Analyze brand preferences by segment
> brand_by_segment <- merged_data %>%
+   group_by(LIFESTAGE, PREMIUM_CUSTOMER, BRAND) %>%
+   summarise(sales = sum(TOT_SALES), .groups = "drop") %>%
+   group_by(LIFESTAGE, PREMIUM_CUSTOMER) %>%
+   mutate(brand_share = sales / sum(sales) * 100) %>%
+   arrange(LIFESTAGE, PREMIUM_CUSTOMER, desc(brand_share))
Error: object 'merged_data' not found
>
> # Analyze pack size preferences by segment
> pack_by_segment <- merged_data %>%
+   group_by(LIFESTAGE, PREMIUM_CUSTOMER, PACK_SIZE) %>%
+   summarise(sales = sum(TOT_SALES), .groups = "drop") %>%
+   group_by(LIFESTAGE, PREMIUM_CUSTOMER) %>%
+   mutate(pack_share = sales / sum(sales) * 100) %>%
+   arrange(LIFESTAGE, PREMIUM_CUSTOMER, desc(pack_share))
Error: object 'merged_data' not found
>
> # --------------------------------------------------
> # 6. CREATE VISUALIZATIONS
> # --------------------------------------------------
> # Plot 1: Top 10 Segments by Total Sales
> top_10_segments <- segment_analysis %>%
+   arrange(desc(total_sales)) %>%
+   slice_head(n = 10) %>%
+   mutate(segment_label = paste(LIFESTAGE, PREMIUM_CUSTOMER, sep
= " - "))
```

```
Error: object 'segment_analysis' not found
>
> ggplot(top_10_segments, aes(x = reorder(segment_label,
total_sales), y = total_sales)) +
+   geom_bar(stat = "identity", fill = "steelblue") +
+   coord_flip() +
+   labs(
+     title = "Top 10 Customer Segments by Total Chip Sales",
+     x = "Customer Segment (Lifestage - Premium)",
+     y = "Total Sales ($)"
+   ) +
+   scale_y_continuous(labels = scales::dollar_format())
Error: object 'top_10_segments' not found
>
> # Plot 2: Brand Preferences for the Top Segment
> # First, find the name of the top segment
> top_segment_name <- top_10_segments$segment_label[1]
Error: object 'top_10_segments' not found
>
> # Get the brand data for just that top segment
> top_segment_brands <- brand_by_segment %>%
+   filter(paste(LIFESTAGE, PREMIUM_CUSTOMER, sep = " - ") ==
top_segment_name) %>%
+   slice_head(n = 5) # Top 5 brands
Error: object 'brand_by_segment' not found
>
> ggplot(top_segment_brands, aes(x = reorder(BRAND, brand_share),
y = brand_share, fill = BRAND)) +
+   geom_bar(stat = "identity") +
+   coord_flip() +
+   labs(
+     title = paste("Top 5 Brands for", top_segment_name),
+     x = "Brand",
+     y = "Share of Spend (%)"
+   ) +
+   theme(legend.position = "none")
Error: object 'top_segment_brands' not found
>
> # --------------------------------------------------
> # 7. SAVE YOUR RESULTS
> # --------------------------------------------------
> # Save the main analysis table to a CSV file
> write.csv(segment_analysis, "customer_segment_analysis.csv",
row.names = FALSE)
Error in eval(expr, p) : object 'segment_analysis' not found
>
> cat("\nAnalysis complete! Your results are saved in
'customer_segment_analysis.csv'.")
```

```
Analysis complete! Your results are saved in
'customer_segment_analysis.csv'.> cat("\nPlots have been generated
in the 'Plots' window.")

Plots have been generated in the 'Plots' window.>
local({fn<-choose.files(filters=Filters[c('R','txt','All'),],index
=4)
+ file.show(fn,header=fn,title='')})
> # --------------------------------------------------
> # 1. LOAD THE DATA
> # --------------------------------------------------
> # Load the transaction and purchase behaviour data
> # Make sure the file names are EXACTLY correct
> transaction_data <- read_excel("QVI Transaction Data.xlsx")
Error: `path` does not exist: 'QVI Transaction Data.xlsx'
> purchase_behaviour <- read_excel("QVI Purchase Behaviour.xlsx")
Error: `path` does not exist: 'QVI Purchase Behaviour.xlsx'
>
> # --------------------------------------------------
> # 2. INITIAL DATA EXPLORATION & CLEANING
> # --------------------------------------------------
> # Look at the first few rows of each dataset
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Error: object 'transaction_data' not found
> head(purchase_behaviour)
Error: object 'purchase_behaviour' not found
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Error: object 'transaction_data' not found
> str(purchase_behaviour)
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> # Check for missing values
> colSums(is.na(transaction_data))
Error: object 'transaction_data' not found
> colSums(is.na(purchase_behaviour))
Error: object 'purchase_behaviour' not found
>
> # Check for and remove any duplicate transactions
> cat("Number of duplicate rows:",
sum(duplicated(transaction_data)), "\n")
Error: object 'transaction_data' not found
> transaction_data <- distinct(transaction_data) # Removes
duplicates
Error: object 'transaction_data' not found
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```

```
> # Convert the DATE column from Excel's number format to a proper
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"1899-12-30")
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> # Extract pack size (the numbers) from the product name
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> # Extract the first word of the product name as the brand
> transaction_data$BRAND <- word(transaction_data$PROD_NAME, 1)
Error: object 'transaction_data' not found
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> # Clean up and standardize some brand names
> transaction_data <- transaction_data %>%
+   mutate(BRAND = case_when(
+     BRAND == "Red" ~ "RRD",
+     BRAND == "Snbts" ~ "Sunbites",
+     BRAND == "Infzns" ~ "Infuzions",
+     BRAND == "WW" ~ "Woolworths",
+     BRAND == "Smith" ~ "Smiths",
+     BRAND == "Dorito" ~ "Doritos",
+     BRAND == "GrnWves" ~ "Grain Waves",
+     BRAND == "Natural" | BRAND == "NCC" ~ "Natural Chip Co",
+     TRUE ~ BRAND
+   ))
Error: object 'transaction_data' not found
>
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> # 4. MERGE THE DATASETS
> # ---------------------------------------------------
> # Combine transaction data with customer data
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Error: object 'transaction_data' not found
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> # ---------------------------------------------------
> # Calculate key metrics for each customer segment
> segment_analysis <- merged_data %>%
+   group_by(LIFESTAGE, PREMIUM_CUSTOMER) %>%
+   summarise(
```

```
+     total_customers = n_distinct(LYLTY_CARD_NBR),
+     total_sales = sum(TOT_SALES),
+     total_transactions = n(),
+     avg_transactions_per_customer = total_transactions /
total_customers,
+     avg_sales_per_customer = total_sales / total_customers,
+     .groups = "drop"
+   ) %>%
+   arrange(desc(total_sales))
Error: object 'merged_data' not found
>
> # Display the top segments by total sales
> print("Top Customer Segments by Total Sales:")
[1] "Top Customer Segments by Total Sales:"
> print(head(segment_analysis, 10))
Error: object 'segment_analysis' not found
>
> # Analyze brand preferences by segment
> brand_by_segment <- merged_data %>%
+   group_by(LIFESTAGE, PREMIUM_CUSTOMER, BRAND) %>%
+   summarise(sales = sum(TOT_SALES), .groups = "drop") %>%
+   group_by(LIFESTAGE, PREMIUM_CUSTOMER) %>%
+   mutate(brand_share = sales / sum(sales) * 100) %>%
+   arrange(LIFESTAGE, PREMIUM_CUSTOMER, desc(brand_share))
Error: object 'merged_data' not found
>
> # Analyze pack size preferences by segment
> pack_by_segment <- merged_data %>%
+   group_by(LIFESTAGE, PREMIUM_CUSTOMER, PACK_SIZE) %>%
+   summarise(sales = sum(TOT_SALES), .groups = "drop") %>%
+   group_by(LIFESTAGE, PREMIUM_CUSTOMER) %>%
+   mutate(pack_share = sales / sum(sales) * 100) %>%
+   arrange(LIFESTAGE, PREMIUM_CUSTOMER, desc(pack_share))
Error: object 'merged_data' not found
>
> # ------------------------------------------------
> # 6. CREATE VISUALIZATIONS
> # ------------------------------------------------
> # Plot 1: Top 10 Segments by Total Sales
> top_10_segments <- segment_analysis %>%
+   arrange(desc(total_sales)) %>%
+   slice_head(n = 10) %>%
+   mutate(segment_label = paste(LIFESTAGE, PREMIUM_CUSTOMER, sep
= " - "))
Error: object 'segment_analysis' not found
>
> ggplot(top_10_segments, aes(x = reorder(segment_label,
total_sales), y = total_sales)) +
```

```
+   geom_bar(stat = "identity", fill = "steelblue") +
+   coord_flip() +
+   labs(
+     title = "Top 10 Customer Segments by Total Chip Sales",
+     x = "Customer Segment (Lifestage - Premium)",
+     y = "Total Sales ($)"
+   ) +
+   scale_y_continuous(labels = scales::dollar_format())
Error: object 'top_10_segments' not found
>
> # Plot 2: Brand Preferences for the Top Segment
> # First, find the name of the top segment
> top_segment_name <- top_10_segments$segment_label[1]
Error: object 'top_10_segments' not found
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Error: object 'brand_by_segment' not found
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> ggplot(top_segment_brands, aes(x = reorder(BRAND, brand_share),
y = brand_share, fill = BRAND)) +
+   geom_bar(stat = "identity") +
+   coord_flip() +
+   labs(
+     title = paste("Top 5 Brands for", top_segment_name),
+     x = "Brand",
+     y = "Share of Spend (%)"
+   ) +
+   theme(legend.position = "none")
Error: object 'top_segment_brands' not found
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> # Save the main analysis table to a CSV file
> write.csv(segment_analysis, "customer_segment_analysis.csv",
row.names = FALSE)
Error in eval(expr, p) : object 'segment_analysis' not found
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> cat("\nAnalysis complete! Your results are saved in
'customer_segment_analysis.csv'.")

Analysis complete! Your results are saved in
'customer_segment_analysis.csv'.> cat("\nPlots have been generated
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```
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> # Load the transaction and purchase behaviour data
> # Make sure the file names are EXACTLY correct
> transaction_data <- read_excel("QVI Transaction Data.xlsx")
Error: `path` does not exist: ‘QVI Transaction Data.xlsx’
> transaction_data <- read_excel("QVI Transaction Data.xlsx")

> # ---------------------------------------------------
> # 1. LOAD THE DATA
> # ---------------------------------------------------
> # Load the transaction and purchase behaviour data
> # Make sure the file names are EXACTLY correct
> transaction_data <- read_excel("QVI Transaction Data.xlsx")
Error: `path` does not exist: ‘QVI Transaction Data.xlsx’
> purchase_behaviour <- read_excel("QVI_purchase_behaviour.xlsx")

> # ---------------------------------------------------
> # 1. LOAD THE DATA
> # ---------------------------------------------------
> # Load the transaction and purchase behaviour data
> # Make sure the file names are EXACTLY correct
> transaction_data <- read_excel("QVI_transaction_data.xlsx")

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> purchase_behaviour <- read_excel("QVI_purchase_behaviour.xlsx")
Error: `path` does not exist: ‘QVI_purchase_behaviour.xlsx’
> # ---------------------------------------------------
> # 1. LOAD THE DATA
> # ---------------------------------------------------
> # Load the transaction and purchase behaviour data
> # Make sure the file names are EXACTLY correct
> transaction_data <- read_excel("QVI Transaction Data.xlsx")
Error: `path` does not exist: ‘QVI Transaction Data.xlsx’
> purchase_behaviour <- read_excel("QVI Purchase Behaviour.xlsx")
Error: `path` does not exist: ‘QVI Purchase Behaviour.xlsx’
> # ---------------------------------------------------
```

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> # 1. LOAD THE DATA
> # ---------------------------------------------------
> # Load the transaction and purchase behaviour data
> # Make sure the file names are EXACTLY correct
> transaction_data <- read_excel("QVI_transaction_data.xlsx")

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> purchase_behaviour <- read_excel("QVI_purchase_behaviour.csv")
Error: Can't establish that the input is either xls or xlsx.
> # ---------------------------------------------------
> # 1. LOAD THE DATA
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```
> # ---------------------------------------------------
> # Load the transaction data (Excel file)
> transaction_data <- read_excel("QVI_transaction_data.xlsx")

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>
> # Load the purchase behaviour data (CSV file)
> # We use read_csv() for .csv files
> purchase_behaviour <- read_csv("QVI_purchase_behaviour.csv")
Error in read_csv("QVI_purchase_behaviour.csv") :
  could not find function "read_csv"
> library(readxl)
> library(dplyr)
> library(ggplot2)
> library(stringr)
```

```
> library(lubridate)
> # ---------------------------------------------------
> # 1. LOAD THE DATA
> # ---------------------------------------------------
> # Load the transaction data (Excel file)
> transaction_data <- read_excel("QVI_transaction_data.xlsx")

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>
> # Load the purchase behaviour data (CSV file)
> purchase_behaviour <- read_csv("QVI_purchase_behaviour.csv")
Error in read_csv("QVI_purchase_behaviour.csv") :
  could not find function "read_csv"
> install.packages("readr")
Installing package into
‘C:/Users/Amooo/AppData/Local/R/win-library/4.5’
(as ‘lib’ is unspecified)
also installing the dependencies ‘bit’, ‘bit64’, ‘clipr’, ‘vroom’,
‘tzdb’
```

```
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/bit_4.6.0.zip
'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/bit64_4.6.0-1
.zip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/clipr_0.8.0.z
ip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/vroom_1.7.0.z
ip'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/tzdb_0.5.0.zi
p'
trying URL
'https://cloud.r-project.org/bin/windows/contrib/4.5/readr_2.2.0.z
ip'
package ‘bit’ successfully unpacked and MD5 sums checked
package ‘bit64’ successfully unpacked and MD5 sums checked
package ‘clipr’ successfully unpacked and MD5 sums checked
package ‘vroom’ successfully unpacked and MD5 sums checked
package ‘tzdb’ successfully unpacked and MD5 sums checked
package ‘readr’ successfully unpacked and MD5 sums checked

The downloaded binary packages are in

C:\Users\Amooo\AppData\Local\Temp\RtmpSGmHag\downloaded_packages
> library(readr)
Warning message:
package ‘readr’ was built under R version 4.5.3
> purchase_behaviour <- read_csv("QVI_purchase_behaviour.csv")

[1mindexing[0m [34mQVI_purchase_behaviour.csv[0m
[=================================================================
-------------------------------------------] [32m262.53MB/s[0m,
eta: [36m 0s[0m
[1mindexing[0m [34mQVI_purchase_behaviour.csv[0m
[=================================================================
===========================================] [32m29.04MB/s[0m,
eta: [36m 0s[0m

Rows: 72637 Columns: 3
── Column specification
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```

```
Delimiter: ","
chr (2): LIFESTAGE, PREMIUM_CUSTOMER
dbl (1): LYLTY_CARD_NBR

i Use `spec()` to retrieve the full column specification for this
data.
i Specify the column types or set `show_col_types = FALSE` to
quiet this message.
> # ---------------------------------------------------
> # 2. INITIAL DATA EXPLORATION & CLEANING
> # ---------------------------------------------------
> # Look at the first few rows of each dataset
> head(transaction_data)
# A tibble: 6 × 8
   DATE STORE_NBR LYLTY_CARD_NBR TXN_ID PROD_NBR PROD_NAME
PROD_QTY TOT_SALES
  <dbl>     <dbl>          <dbl>  <dbl>    <dbl> <chr>
<dbl>     <dbl>
1 43390         1           1000      1        5 Natural Chip
Compny SeaSalt175g          2       6
2 43599         1           1307    348       66 CCs Nacho Cheese
175g                        3       6.3
3 43605         1           1343    383       61 Smiths Crinkle
Cut  Chips Chicken 170g          2       2.9
4 43329         2           2373    974       69 Smiths Chip
Thinly  S/Cream&Onion 175g          5      15
5 43330         2           2426   1038      108 Kettle Tortilla
ChpsHny&Jlpno Chili 150g        3      13.8
6 43604         4           4074   2982       57 Old El Paso Salsa
Dip Tomato Mild 300g        1       5.1
> head(purchase_behaviour)
# A tibble: 6 × 3
  LYLTY_CARD_NBR LIFESTAGE              PREMIUM_CUSTOMER
           <dbl> <chr>                  <chr>
1           1000 YOUNG SINGLES/COUPLES  Premium
2           1002 YOUNG SINGLES/COUPLES  Mainstream
3           1003 YOUNG FAMILIES         Budget
4           1004 OLDER SINGLES/COUPLES  Mainstream
5           1005 MIDAGE SINGLES/COUPLES Mainstream
6           1007 YOUNG SINGLES/COUPLES  Budget
>
> # Check the structure (column names and types)
> str(transaction_data)
tibble [264,836 × 8] (S3: tbl_df/tbl/data.frame)
 $ DATE          : num [1:264836] 43390 43599 43605 43329 43330
...
 $ STORE_NBR     : num [1:264836] 1 1 1 2 2 4 4 4 5 7 ...
 $ LYLTY_CARD_NBR: num [1:264836] 1000 1307 1343 2373 2426 ...
```

```
 $ TXN_ID        : num [1:264836] 1 348 383 974 1038 ...
 $ PROD_NBR      : num [1:264836] 5 66 61 69 108 57 16 24 42 52
...
 $ PROD_NAME     : chr [1:264836] "Natural Chip        Compny
SeaSalt175g" "CCs Nacho Cheese    175g" "Smiths Crinkle Cut  Chips
Chicken 170g" "Smiths Chip Thinly  S/Cream&Onion 175g" ...
 $ PROD_QTY      : num [1:264836] 2 3 2 5 3 1 1 1 1 2 ...
 $ TOT_SALES     : num [1:264836] 6 6.3 2.9 15 13.8 5.1 5.7 3.6
3.9 7.2 ...
> str(purchase_behaviour)
spc_tbl_ [72,637 × 3] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ LYLTY_CARD_NBR  : num [1:72637] 1000 1002 1003 1004 1005 ...
 $ LIFESTAGE       : chr [1:72637] "YOUNG SINGLES/COUPLES" "YOUNG
SINGLES/COUPLES" "YOUNG FAMILIES" "OLDER SINGLES/COUPLES" ...
 $ PREMIUM_CUSTOMER: chr [1:72637] "Premium" "Mainstream" "Budget"
"Mainstream" ...
 - attr(*, "spec")=
  .. cols(
  ..   LYLTY_CARD_NBR = col_double(),
  ..   LIFESTAGE = col_character(),
  ..   PREMIUM_CUSTOMER = col_character()
  .. )
 - attr(*, "problems")=<externalptr>
>
> # Check for missing values
> colSums(is.na(transaction_data))
          DATE      STORE_NBR LYLTY_CARD_NBR         TXN_ID
PROD_NBR      PROD_NAME       PROD_QTY      TOT_SALES
             0              0              0              0
0              0              0              0
> colSums(is.na(purchase_behaviour))
  LYLTY_CARD_NBR        LIFESTAGE PREMIUM_CUSTOMER
               0                0                0
>
> # Check for and remove any duplicate transactions
> cat("Number of duplicate rows:",
sum(duplicated(transaction_data)), "\n")
Number of duplicate rows: 1
> transaction_data <- distinct(transaction_data) # Removes
duplicates
>
> # Convert the DATE column from Excel's number format to a proper
date
> transaction_data$DATE <- as.Date(transaction_data$DATE, origin =
"1899-12-30")
> # ---------------------------------------------------
> # 3. FEATURE ENGINEERING
> # ---------------------------------------------------
```

```
> # Extract pack size (the numbers) from the product name
> transaction_data$PACK_SIZE <-
as.numeric(str_extract(transaction_data$PROD_NAME, "\\d+"))
>
> # Extract the first word of the product name as the brand
> transaction_data$BRAND <- word(transaction_data$PROD_NAME, 1)
>
> # Clean up and standardize some brand names
> transaction_data <- transaction_data %>%
+   mutate(BRAND = case_when(
+     BRAND == "Red" ~ "RRD",
+     BRAND == "Snbts" ~ "Sunbites",
+     BRAND == "Infzns" ~ "Infuzions",
+     BRAND == "WW" ~ "Woolworths",
+     BRAND == "Smith" ~ "Smiths",
+     BRAND == "Dorito" ~ "Doritos",
+     BRAND == "GrnWves" ~ "Grain Waves",
+     BRAND == "Natural" | BRAND == "NCC" ~ "Natural Chip Co",
+     TRUE ~ BRAND
+   ))
> # ---------------------------------------------------
> # 4. MERGE THE DATASETS
> # ---------------------------------------------------
> # Combine transaction data with customer data
> merged_data <- transaction_data %>%
+   left_join(purchase_behaviour, by = "LYLTY_CARD_NBR")
>
> # ---------------------------------------------------
> # 5. ANALYZE CUSTOMER SEGMENTS
> # ---------------------------------------------------
> # Calculate key metrics for each customer segment
> segment_analysis <- merged_data %>%
+   group_by(LIFESTAGE, PREMIUM_CUSTOMER) %>%
+   summarise(
+     total_customers = n_distinct(LYLTY_CARD_NBR),
+     total_sales = sum(TOT_SALES),
+     total_transactions = n(),
+     avg_transactions_per_customer = total_transactions /
total_customers,
+     avg_sales_per_customer = total_sales / total_customers,
+     .groups = "drop"
+   ) %>%
+   arrange(desc(total_sales))
>
> # Display the top segments by total sales
> print("Top Customer Segments by Total Sales:")
[1] "Top Customer Segments by Total Sales:"
> print(head(segment_analysis, 10))
```

```
# A tibble: 10 × 7
   LIFESTAGE             PREMIUM_CUSTOMER total_customers
total_sales total_transactions avg_transactions_per_customer
avg_sales_per_customer
   <chr>                 <chr>                       <int>
<dbl>              <int>                          <dbl>
<dbl>
 1 OLDER FAMILIES        Budget                       4675
168363.              23160                           4.95
36.0
 2 YOUNG SINGLES/COUPLES Mainstream                   8088
157622.              20854                           2.58
19.5
 3 RETIREES              Mainstream                   6479
155677.              21466                           3.31
24.0
 4 YOUNG FAMILIES        Budget                       4017
139346.              19122                           4.76
34.7
 5 OLDER SINGLES/COUPLES Budget                       4929
136770.              18407                           3.73
27.7
 6 OLDER SINGLES/COUPLES Mainstream                   4930
133394.              18318                           3.72
27.1
 7 OLDER SINGLES/COUPLES Premium                      4750
132257.              17753                           3.74
27.8
 8 RETIREES              Budget                       4454
113148.              15201                           3.41
25.4
 9 OLDER FAMILIES        Mainstream                   2831
103446.              14244                           5.03
36.5
10 RETIREES              Premium                      3872
97646.              13096                           3.38
25.2
> # ----------------------------------------------
> # 6. CREATE VISUALIZATIONS
> # ----------------------------------------------
> # Plot 1: Top 10 Segments by Total Sales
> top_10_segments <- segment_analysis %>%
+   arrange(desc(total_sales)) %>%
+   slice_head(n = 10) %>%
+   mutate(segment_label = paste(LIFESTAGE, PREMIUM_CUSTOMER, sep
= " - "))
>
```

```
> ggplot(top_10_segments, aes(x = reorder(segment_label,
total_sales), y = total_sales)) +
+   geom_bar(stat = "identity", fill = "steelblue") +
+   coord_flip() +
+   labs(
+     title = "Top 10 Customer Segments by Total Chip Sales",
+     x = "Customer Segment (Lifestage - Premium)",
+     y = "Total Sales ($)"
+   ) +
+   scale_y_continuous(labels = scales::dollar_format())
>
> # --------------------------------------------------
> # 7. SAVE YOUR RESULTS
> # --------------------------------------------------
> # Save the main analysis table to a CSV file
> write.csv(segment_analysis, "customer_segment_analysis.csv",
row.names = FALSE)
>
> cat("\nAnalysis complete! Your results are saved in
'customer_segment_analysis.csv'.")

Analysis complete! Your results are saved in
'customer_segment_analysis.csv'.> cat("\nA plot should have
appeared in a new window.")
```

Initial Findings Summary

The analysis of the QVI transaction and purchase behaviour data revealed key insights into customer segments and their chip purchasing habits.

1. Top Customer Segments by Total Sales:
The data shows that the highest total sales come from the following segments:

- OLDER FAMILIES - Budget
- YOUNG SINGLES/COUPLES - Mainstream
- RETIREES - Mainstream

These three segments are the primary drivers of revenue for the chip category, with OLDER FAMILIES (Budget) leading significantly.

2. Key Observations:

- A total of 1 duplicate transaction was found and removed to ensure data accuracy.

- The analysis successfully derived new features, including pack size and brand name, from the product description to enable deeper analysis.
- While the YOUNG SINGLES/COUPLES - Mainstream segment has the second-highest total sales, it has a very low average sales per customer ($19.5) compared to top segments like OLDER FAMILIES (Budget) and YOUNG FAMILIES (Budget), who spend over $34 per customer. This suggests the Mainstream segment makes more frequent, smaller purchases.
- The OLDER FAMILIES - Mainstream segment, despite having fewer customers than many others, has a high average spend per customer ($36.5) and a high number of transactions per customer (5.03), indicating they are highly valuable and engaged shoppers.

3. Initial Strategic Implications:

- Marketing efforts should prioritize the top-spending segments, particularly OLDER FAMILIES (both Budget and Mainstream) and YOUNG FAMILIES - Budget.
- For the large YOUNG SINGLES/COUPLES - Mainstream segment, promotional strategies could focus on increasing basket size (e.g., multi-buy deals like "buy 2 for $5") to lift their average spend.
- The high value of the OLDER FAMILIES and RETIREE segments suggests they are receptive to category-specific promotions and should be targeted with loyalty offers.